



100081

北京市海淀区彩和坊路 8 号 7 层 715-716 室 北京天悦专利代理事务所
(普通合伙)
田明(010-84934084) 任晓航(010-84934084)

发文日:

2024 年 02 月 26 日



申请号: 202111185786.5

发文序号: 2024022601553130

申请人: 华北电力大学

发明创造名称: 一种钙钛矿填充微孔面板形成的微结构闪烁屏及制备方法

第一次审查意见通知书

1. ☒ 应申请人提出的实质审查请求, 根据专利法第 35 条第 1 款的规定, 国家知识产权局对上述发明专利申请进行实质审查。

☐ 根据专利法第 35 条第 2 款的规定, 国家知识产权局决定自行对上述发明专利申请进行审查。

2. ☐ 申请人要求以其在:

☐ 申请人已经提交了经原受理机构证明的第一次提出的在先申请文件的副本。

☐ 申请人尚未提交经原受理机构证明的第一次提出的在先申请文件的副本, 根据专利法第 30 条的规定视为未要求优先权要求。

3. ☐ 经审查, 申请人于_____提交的修改文件, 不符合专利法实施细则第 57 条第 1 款的规定, 不予接受。

4. 审查针对的申请文件:

☒ 原始申请文件。 ☐ 分案申请递交日提交的文件。 ☐ 下列申请文件:

5. ☐ 本通知书是在未进行检索的情况下作出的。

☒ 本通知书是在进行了检索的情况下作出的。

☒ 本通知书引用下列对比文件(其编号在今后的审查过程中继续沿用):

编号	文件号或名称	公开日期 (或抵触申请的申请日)
1	“Giant Optical Anisotropy of Perovskite Nanowire Array Films”, Chun Ho Lin 等, 《Advanced Functional Materials》, 第 30 卷, 第 1-7 页	2020-02-19
2	“基于微孔硅阵列的像素化 γ -CuI 闪烁转换屏的研制及性能表征”, 孙寿强 等, 《光谱学与光谱分析》, 第 37 卷第 5 期, 第 1566-1570 页	2017-05-15

6. 审查的结论性意见:

关于说明书:

☐ 申请的内容属于专利法第 5 条规定的不授予专利权的范围。

☐ 说明书不符合专利法第 26 条第 3 款的规定。

☐ 说明书不符合专利法第 33 条的规定。

☐ 说明书的撰写不符合专利法实施细则第 20 条的规定。



国家知识产权局

☐ _____

关于权利要求书：

- ☐ 权利要求 _____ 不符合专利法第 2 条第 2 款的规定。
- ☐ 权利要求 _____ 不符合专利法第 9 条第 1 款的规定。
- ☐ 权利要求 _____ 不具备专利法第 22 条第 2 款规定的新颖性。
- ☒ 权利要求 1-2 不具备专利法第 22 条第 3 款规定的创造性。
- ☐ 权利要求 _____ 不具备专利法第 22 条第 4 款规定的实用性。
- ☐ 权利要求 _____ 属于专利法第 25 条规定的不授予专利权的范围。
- ☒ 权利要求 2 和 4 不符合专利法第 26 条第 4 款的规定。
- ☐ 权利要求 _____ 不符合专利法第 31 条第 1 款的规定。
- ☐ 权利要求 _____ 不符合专利法第 33 条的规定。
- ☐ 权利要求 _____ 不符合专利法实施细则第 22 条的规定。
- ☐ 权利要求 _____ 不符合专利法实施细则第 23 条的规定。
- ☐ 权利要求 _____ 不符合专利法实施细则第 24 条的规定。
- ☐ 权利要求 _____ 不符合专利法实施细则第 25 条的规定。
- ☐ _____

- ☐ 申请不符合专利法第 26 条第 5 款或者实施细则第 29 条的规定。
- ☐ 申请不符合专利法第 19 条第 1 款的规定。
- ☐ 申请不符合专利法实施细则第 11 条的规定。
- ☐ 分案申请不符合专利法实施细则第 49 条第 1 款的规定。

上述结论性意见的具体分析见本通知书的正文部分。

7. 基于上述结论性意见，审查员认为：

- ☐ 申请人应当按照通知书正文部分提出的要求，对申请文件进行修改。
- ☒ 申请人应当在意见陈述书中论述其专利申请可以被授予专利权的理由，并对通知书正文部分中指出的不符合规定之处进行修改，否则将不能授予专利权。
- ☐ 专利申请中没有可以被授予专利权的实质性内容，如果申请人没有陈述理由或者陈述理由不充分，其申请将被驳回。

☐ _____

8. 申请人应注意下列事项：

- (1) 根据专利法第 37 条的规定，申请人应在收到本通知书之日起的 4 个月内陈述意见，如果申请人无正当理由逾期不答复，其申请被视为撤回。
- (2) 申请人对其申请的修改应当符合专利法第 33 条的规定，不得超出原说明书和权利要求书记载的范围，同时申请人对专利申请文件进行的修改应当符合专利法实施细则第 57 条第 3 款的规定，按照本通知书的要求进行修改。
- (3) 申请人的意见陈述书和/或修改文本应邮寄或递交国家知识产权局专利局受理处，凡未邮寄或递交给受理处的文件不具备法律效力。
- (4) 未经预约，申请人和/或代理师不得前来国家知识产权局专利局与审查员举行会晤。
- (5) 对进入实质审查阶段的发明专利申请，在第一次审查意见通知书答复期限届满前（已提交答复意见的除外），主动申请撤回的，可以请求退还 50% 的专利申请实质审查费。

9. 本通知书正文部分共有 2 页，并附有下列附件：

- ☒ 引用的对比文件的复印件共 2 份 13 页。
- ☐ _____

审查员：马梨

联系电话：028-62967973

审查部门：专利审查协作四川中心



210401
2023.03

纸件申请，回函请寄：100088 北京市海淀区蓟门桥西土城路 6 号 国家知识产权局专利局受理处收
电子申请，应当通过电子专利申请系统以电子文件形式提交相关文件。除另有规定外，以纸件等其他形式提交的文件视为未提交。



第一次审查意见通知书

申请号:2021111857865

本发明涉及一种钙钛矿填充微孔面板形成的微结构闪烁屏及制备方法,经审查,现提出如下审查意见。

1、权利要求1不符合专利法第22条第3款有关创造性的规定

权利要求1要求保护一种钙钛矿填充微孔面板形成的微结构闪烁屏。对比文件1(“Giant Optical Anisotropy of Perovskite Nanowire Array Films”, Chun Ho Lin 等,《Advanced Functional Materials》,第30卷,第1-7页,2020年2月19日)为最接近的现有技术,其公开了一种钙钛矿填充微孔面板形成的微结构闪烁屏,并具体公开了:首先,我们将制备好的 $\text{CH}_3\text{NH}_3\text{PbBr}_3$ (相当于钙钛矿材料的闪烁体)油墨喷墨印刷在纳米多孔AAO(相当于微孔面板)基底上,并对AAO的底侧施加真空泵,以将油墨引导到纳米孔中(相当于所述钙钛矿材料的闪烁体均匀填充每个微孔)。注意,如果真空抽吸太强,可能会出现钙钛矿油墨的不连续分布,导致NW的质量较差。因此,应小心地将真空流量控制在中等水平。将油墨引导到纳米孔中后,在 90°C 的热板上对样品进行热退火,以蒸发不需要的溶剂。然后,通过纳米多孔AAO结构将钙钛矿重结晶为NW形状,由图1可以知道AAO的纳米多孔规则阵列分布(相当于所述微孔面板具有规则的微孔阵列结构)。

该权利要求所保护的技术方案与对比文件1公开的技术内容相比,其区别技术特征在于:微孔孔壁内部包括光吸收层,微孔内壁为光反射层。基于上述区别技术特征可以确定权利要求1相对于对比文件1实际解决的技术问题是“提升X射线成像的空间分辨率”。

针对于区别技术特征,对比文件2(“基于微孔硅阵列的像素化 $\gamma\text{-CuI}$ 闪烁转换屏的研制及性能表征”,孙寿强等,《光谱学与光谱分析》,第37卷第5期,第1566-1570页,2017年5月31日)公开了(参见第1567页):图2(a)和(b)是高温氧化后具有 SiO_2 反射层结构的微孔硅阵列模板的表面、断面SEM形貌图,可以看到,模板由周期性均匀分布的微孔组成,微孔排列周期(孔间距)为 $4\mu\text{m}$ 、孔径约为 $2.5\mu\text{m}$ 、壁厚约为 $1.5\mu\text{m}$,其中孔壁表面 SiO_2 层厚度约为500nm,如图2(c),具有 SiO_2 氧化层的微孔硅阵列,可同时利用界面的全反射和孔壁中间的硅层对可见光的吸收(相当于微孔孔壁内部包括光吸收层,微孔内壁为光反射层),抑制闪烁光的横向传播如图2(d)。可见,上述技术特征在对比文件2中所起的作用与其在本申请中所起的相同,均是提升X射线成像的空间分辨率。

由此可知,在对比文件1的基础上结合对比文件2,得出该权利要求的技术方案,对本技术领域的人员来说是显而易见的,因此该权利要求所要求保护的技术方案不具有突出的实质性特点和显著的进步,因而不具备专利法第22条第3款规定的创造性。

2、权利要求2不符合专利法第22条第3款有关创造性的规定

权利要求2对权利要求1做了进一步限定,对比文件1公开了:卤化物钙钛矿 APbX_3 ($\text{A} = \text{MA}, \text{FA}, \text{and Cs}$; $\text{X} = \text{Cl}, \text{Br}, \text{and I}$)(相当于“所述钙钛矿材料的闪烁体包括但不限于以下种类的钙钛矿:三维钙钛矿 ABX_3 、二维钙钛矿 $\text{A}_2\text{B}'\text{B}''\text{X}_6$ 、一维钙钛矿 A_3BX_5 以及零维钙钛矿 A_4BX_6 ;其中,所述A选自铯离子 Cs^+ 、甲脒阳离子 FA^+ 、甲铵阳离子 MA^+ 、正辛铵阳离子 OA^+ 、苄乙铵阳离子 PEA^+ 、苄甲铵阳离子 PMA^+ 、叔丁铵阳离子中的一种或几种;所述B选自 Pb^{2+} 、 Sn^{2+} 中的一种或几种,所述B'选自 Ag^+ 、 Na^+ 、 Cu^+ 中的一种或几种;所述B''选自 Bi^{3+} 、 Sb^{3+} 、 In^{3+} 中的一种或几种;所述X选自 I^- 、 Br^- 、 Cl^- 中的一种或几种”)。

因此,在其所引用的权利要求不具备创造性的情况下,该权利要求也不具备创造性。

3、权利要求2和4不符合专利法第26条第4款的规定

权利要求2和4中的“但不限于”导致权利要求保护范围的不清楚,因此,权利要求2和4不符合专利法第26条第4款的规定。

申请人应在本通知书指定的答复期限内作出答复,对本通知书中提出的所有问题作出说明,并依据需要对专利申请文件作出修改,并在意见陈述书中论述新修改的独立权利要求符合专利法及其实施细则规定的理由。申请人对申请文件的修改应当符合专利法第三十三条的规定,不得超出原说明书和权利要求书记载的范围。



审查员姓名:马梨
审查员代码:30141043



国家知识产权局

检索报告

申请号：2021111857865	申请日：2021 年 10 月 12 日	首次检索			
申请人：华北电力大学	最早的优先权日：				
权利要求项数：10	说明书段数：61+6				
审查员确定的 IPC 分类号：G01T 1/202					
检索记录信息：US2003178574A1: 34 WPABS,ENTXT, perovskite AND scintillat+ AND (screen or imag+) and (microhole or microwell or micropor+ or (micro+ 3w (channel or passage))) CN209182517U: 26 CNTXT,CJFD, (张小东 or 王翔)/in and 华北电力大学/pa KR20200119141A: 64 VCN,CJFD, (钙钛矿 and 闪烁 and (孔 or 洞)) and (语义限定结果 1) 其中： 语义限定结果 1 为: CNTXT, ENTXTC, CJFD, 语义限定, 语义基准: 2021111857865; DE102014212424A1: 64 VCN,CJFD, (钙钛矿 and 闪烁 and (孔 or 洞)) and (语义限定结果 1) 其中： 语义限定结果 1 为: CNTXT, ENTXTC, CJFD, 语义限定, 语义基准: 2021111857865; WO2021077006A1: 7 VCN, G01T1/iC AND (钙钛矿 s (微孔 or 多孔 or 细孔)) CN108351428A: 7 VCN, G01T1/iC AND (钙钛矿 s (微孔 or 多孔 or 细孔)) US2014342236A1: 3 WPABS,ENTXT, perovskite AND scintillat+ AND AAO JP2018096714A: 163 WPABS,ENTXT, G01T1/IC AND perovskite AND scintillat+ AND (screen or imag+) and (hole or opening or aperture or pores or porus) US2017160405A1: 163 WPABS,ENTXT, G01T1/IC AND perovskite AND scintillat+ AND (screen or imag+) and (hole or opening or aperture or pores or porus) CN113299672A: 8 VCN, 闪烁体 AND (钙钛矿 s (微孔 or 多孔 or 细孔)) and (成像 or 显示 or 屏) Giant Optical Anisotropy of Perovskite Nanowire Array Films: 谷歌学术, perovskite scintillation screen microhole 基于微孔硅阵列的像素化 γ-CuI 闪烁转换屏的研制及性能表征: CJFD 语义检索, 语义基 准: 2021111857865 212 CNKI 高级检索, (闪烁体 * 钙钛矿 * (多孔 + 微孔))/全文(精确)					
相 关 专 利 文 献					
类型	国别以及代码[11] 给出的文献号	代码[43]或[45] 给出的日期	IPC 分类号	相关的段落 和 / 或图号	涉及的权 利要求



国家知识产权局

A	US2003178574A1	2003-09-25	G01T3/06	全文	1-10
A	CN209182517U	2019-07-30	G01T3/06	全文	1-10
A	KR20200119141A	2020-10-19	H01L33/36	全文	1-10
A	DE102014212424A1	2015-06-18	H01L31/115	全文	1-10
A	WO2021077006A1	2021-04-22	B32B27/18	全文	1-10
A	CN108351428A	2018-07-31	G01T1/24	全文	1-10
A	US2014342236A1	2014-11-20	H01M4/04	全文	1-10
A	JP2018096714A	2018-06-21	G01T1/20	全文	1-10
A	US2017160405A1	2017-06-08	G01T1/20	全文	1-10
A	CN113299672A	2021-08-24	H01L27/144	全文	1-10

相 关 非 专 利 文 献					
类型	书名（包括版本号和卷号）	出版日期	作者姓名和出版者名称	相关页数	涉及的权利要求
类型	期刊或文摘名称 （包括卷号和期号）	发行日期	作者姓名和文章标题	相关页数	涉及的权利要求



国家知识产权局

Y	Advanced Functional Materials, 第 30 卷	2020-02-19	Chun Ho Lin 等,Giant Optical Anisotropy of Perovskite Nanowire Array Films	第 1-7 页	1-2
Y	光谱学与光谱分析,05	2017-05-15	孙寿强;顾牡;陈婷婷;张娟 楠;刘小林;刘波;黄世明; 倪晨,,基于微孔硅阵列的 像素化 γ -CuI 闪烁转换 屏的研制及性能表征	第 1566-157 0 页	1-2
类型	网址	网络发布日 或公开日	作者姓名和网页标题	相关部分	涉及的权利要求

表格填写说明事项:

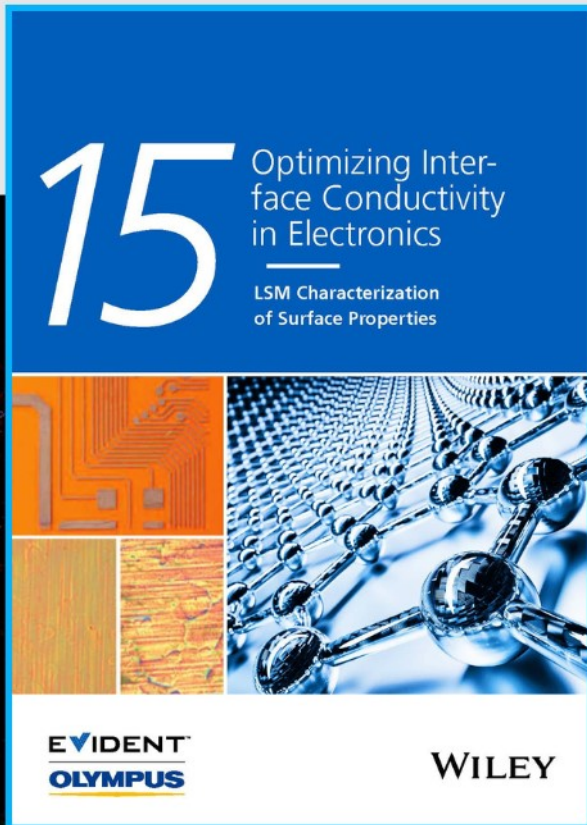
1. 审查员实际检索领域的 IPC 分类号应当填写到大组和 / 或小组所在的分类位置。
2. 期刊或其它定期出版物的名称可以使用符合一般公认的国际惯例的缩写名称。
3. 相关文件的类型说明:
X: 单独影响权利要求的新颖性或创造性的文件;
Y: 与本检索报告中其他 Y 类文件组合后影响权利要求的创造性的文件;
A: 背景技术文件, 即反映权利要求的部分技术特征或者有关的现有技术的文件;
R: 任何单位或个人在申请日向专利局提交的、属于同样的发明创造的专利或专利申请文件。
P: 中间文件, 其公开日在申请的申请日与所要求的优先权日之间的文件, 或者会导致需要核实该申请优先权的文件;
E: 单独影响权利要求新颖性的抵触申请文件;
T: 申请日或优先权日当天或之后公布的, 可以对所要求保护发明的理论或原理提供清楚解释的文件, 或者可显示出所要求保护发明的推理或事实不成立的文件;
L: 除 X、Y、A、R、P、E 和 T 类文件之外的文件。

审 查 员: 马梨
2024 年 02 月 23 日

审查部门: 专利审查协作四川中心



Optimizing Interface Conductivity in Electronics



The latest eBook from
Advanced Optical Metrology.
Download for free.

Surface roughness is a key parameter for judging the performance of a given material's surface quality for its electronic application. A powerful tool to measure surface roughness is 3D laser scanning confocal microscopy (LSM), which will allow you to assess roughness and compare production and finishing methods, and improve these methods based on mathematical models.

Focus on creating high-conductivity electronic devices with minimal power loss using laser scanning microscopy is an effective tool to discern a variety of roughness parameters.

EVIDENT
OLYMPUS

WILEY

Giant Optical Anisotropy of Perovskite Nanowire Array Films

Chun-Ho Lin, Chieh-Yu Kang, Ting-Zhu Wu, Chun-Lin Tsai, Chin-Wei Sher, Xinwei Guan, Po-Tsung Lee, Tom Wu, Chih-Hsiang Ho,* Hao-Chung Kuo,* and Jr-Hau He*

Polarizers play a key role in generating polarized light for display, imaging, and data communication, but adoption often suffers from high optical loss. Recently, due to superior optoelectronic properties, halide perovskites have been widely developed for lighting applications; however, highly polarized emission (polarization degree >0.8) has not yet been realized with perovskites. Herein, by incorporating inkjet printing and an anodic aluminum oxide (AAO) confinement strategy, highly ordered perovskite nanowire (NW) arrays are demonstrated for anisotropic optical applications. The optical device based on perovskite NW arrays reveals a high photoluminescence external quantum efficiency of 21.6% and emits highly polarized light with polarization degree up to 0.84. The highly polarized emission from perovskite NW arrays has potential to considerably reduce the optical loss of polarizers, which may attract great interest in developing polarized light sources for next-generation optoelectronic applications.

Although human eye lacks the ability to distinguish polarized light, it plays a vital role in connecting human lives with advanced optoelectronics technology, including liquid crystal display (LCD) backlights,^[1,2] visible light communication,^[3,4] ultrasensitive photodetectors,^[5,6] and optical quantum computing.^[7,8] For example, in LCD panels, polarizer is a key component to convert backlight emission into polarized light. However, it inevitably causes huge optical loss ≈50% during the light conversion in conventional LCD.^[2] To address this issue, several strategies have been proposed to create polarized backlight, such as dielectric confinement of optical electric field,^[9] anisotropic effective transition dipole moments,^[10] and the splitting of the exciton fine structure.^[11]

1. Introduction

Polarized light is a light wave with electric field restricted in a certain plane perpendicular to the direction of propagation.

The realization of polarized backlight can significantly increase the light transmittance through the polarizer, thus improving the efficiency of the LCD.

Halide perovskites with the general formula of APbX_3 ($A = \text{MA}$, FA , and Cs ; $X = \text{Cl}$, Br , and I) have been widely adopted in photovoltaics and lighting applications owing to their high absorption coefficient,^[12] intense photoluminescence (PL) emission,^[13] controllable optoelectronic property,^[14] and ability to grow by cost-effective solution process.^[15] With elongated 1D structure and high refractive index compared with the environment, perovskite nanowires (NWs) and nanorods are able to emit strongly polarized light.^[16–19] For instance, Tauber et al. reported the polarized emission from an individual MAPbI_3 NW with high degree of linear polarization (DOLP) of 0.7.^[20] However, the anisotropy polarization will be dramatically reduced for NW clusters due to random distribution and disordered arrangement of NWs, which is not practical for the scalable light emission application. As a proof of concept, Güner et al. encapsulated CsPbBr_3 NWs into electrospun polyurethane (PU) fibers, but the emitted light from the specimen only showed a DOLP of 0.3.^[21]

Generally, to realize highly polarized emission, NWs need to be aligned in ordered arrangement.^[22] Electrospinning, mechanical rubbing, anisotropic etching, and lithography patterning are commonly used methods to fabricate well aligned nanostructures.^[23,24] However, the mentioned strategies are not applicable for ionic perovskites because these materials are unstable and tend to be degraded when they contact with polar compounds or environment.^[25,26] For example, a recent report demonstrated that in typical lithography process, both

Dr. C.-H. Lin, X. Guan, Prof. T. Wu
School of Materials Science and Engineering
University of New South Wales (UNSW)
Sydney, NSW 2052, Australia

C.-Y. Kang, Dr. T.-Z. Wu, C.-L. Tsai, Dr. C.-W. Sher,
Prof. P.-T. Lee, Prof. H.-C. Kuo
Department of Photonics and Institute of Electro-Optical Engineering
College of Electrical and Computer Engineering
National Chiao Tung University
Hsinchu 30010, Taiwan
E-mail: hckuo@faculty.nctu.edu.tw

Dr. T.-Z. Wu
Department of Electronic Science
Fujian Engineering Research Center for Solid-State Lighting
Xiamen University
Xiamen 361005, China

Dr. C.-H. Ho
Raysolution LLC
San Jose, CA 95129, USA
E-mail: sam.chihhsiang.ho@gmail.com

Prof. J.-H. He
Department of Materials Science and Engineering
City University of Hong Kong
Kowloon, Hong Kong SAR 999077, China
E-mail: jrhaue@cityu.edu.hk

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adfm.201909275>.

DOI: 10.1002/adfm.201909275

3D and 2D perovskites were damaged by polar solvent in just a few seconds.^[27] Besides the difficulty of aligning perovskite NWs, substrate and other supporting materials should exhibit high transmittance to reduce the optical loss during excited light emission. For these reasons, while high DOLP >0.8 was reported based on other semiconductors,^[28,29] highly polarized emission from perovskite films has not yet achieved.

Inkjet printing is regarded as the forefront of microelectronics fabrication technology due to its simple, flexible, rapid, and cost-effective manufacturing approach, which provides an efficient way for mass production.^[30] In this work, by incorporating inkjet printing technique and nanoporous anodic aluminum oxide (AAO) confinement strategy, we realized the anisotropic growth of perovskite and successfully fabricated high quality and orientationally aligned perovskite NW array films for extremely polarized emission. The AAO displays high transmittance >70%, thereby allowing optical excitation and light emission of perovskite NWs. Using blue GaN light-emitting diodes (LEDs) as optical pumping source, the printed $\text{CH}_3\text{NH}_3\text{PbBr}_3$ NW arrays demonstrate intense emission featuring peak wavelength of 540 nm, narrow full width at half maximum (FWHM) of 22 nm, PL external quantum efficiency (EQE) of 21.6%, and DOLP up to 0.84. With excellent performance of polarized emissions and efficiency, the perovskite NW array films are promising for polarized light applications.

2. Results and Discussion

The fabrication flowchart of perovskite NW array films is shown in **Figure 1a**. First, we inkjet-printed as-prepared $\text{CH}_3\text{NH}_3\text{PbBr}_3$ inks on a nanoporous AAO substrate and applied a vacuum pumping to the bottom side of the AAO to guide the inks into the nanopores. Note that if the vacuum

suction is too strong, a discontinuous distribution of perovskite inks could occur, leading to inferior quality of NWs. Therefore, the vacuum flow should be carefully controlled in a moderate level. After the inks were guided into the nanopores, the sample was thermal annealed on a hotplate at 90 °C to evaporate the unwanted solvents. Then, the perovskites were recrystallized to the NW shapes templated by the nanoporous AAO structure. Because the nanopores of AAO are oriented along the through-plane of AAO, the resulting perovskite NWs were arranged in a consistent orientation (**Figure 1b**). In this work, we chose $\text{CH}_3\text{NH}_3\text{PbBr}_3$ as the active material because it presents stronger PL efficiency than chloride and iodide perovskite counterparts. **Figure 1c** demonstrates the cross-sectional scanning electron microscopy (SEM) image of $\text{CH}_3\text{NH}_3\text{PbBr}_3$ perovskite NWs within the AAO nanopores. Note that the perovskite NWs are well aligned along the through-plane of AAO substrate, thus being expected to enable highly anisotropic polarization effect.

Figure 2a illustrates the through-plane design of the optical device. We utilized a 450 nm GaN LED at the bottom of $\text{CH}_3\text{NH}_3\text{PbBr}_3$ NW/AAO film to pump the perovskite NW array film optically from the through-plane orientation. Then, the top surface of AAO was covered by a distributed Bragg reflector (DBR) in the blue wavelengths, which plays the role to reuse the blue light and considerably enhance photoexcitation on the perovskite. Before characterizing the optical device, we undertook X-ray diffraction (XRD), PL, and ultraviolet–visible (UV–vis) absorption measurements to verify the crystallinity and optical properties of $\text{CH}_3\text{NH}_3\text{PbBr}_3$ NW arrays. **Figure 2b** presents the XRD pattern of $\text{CH}_3\text{NH}_3\text{PbBr}_3$ NW/AAO sample. We can see that all diffraction peak positions are in good agreement with the cubic $\text{CH}_3\text{NH}_3\text{PbBr}_3$ phase.^[31] Additionally, the line width of diffraction peaks is very small, confirming the high crystalline quality of as-obtained perovskite NW arrays.

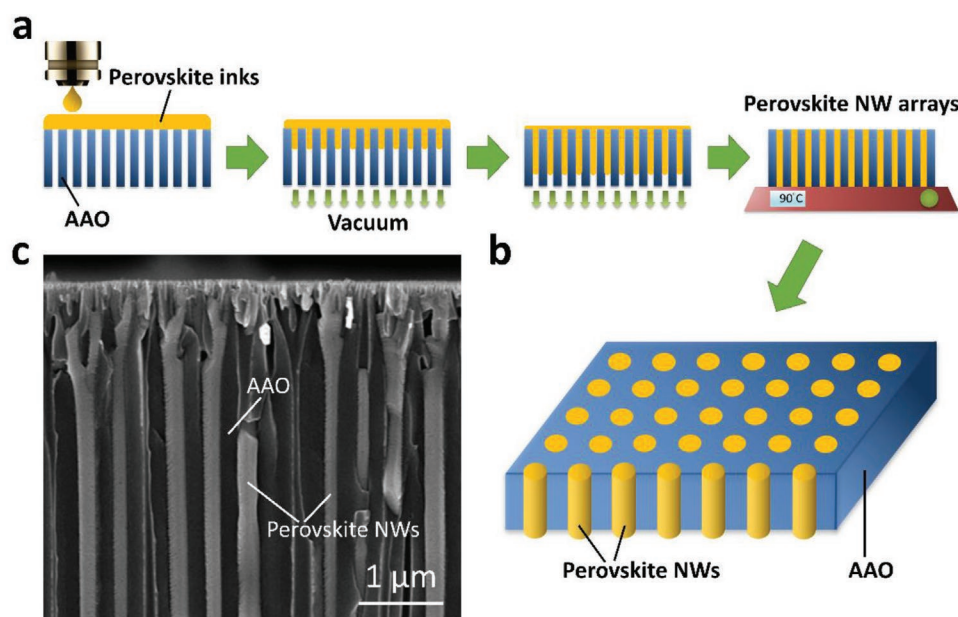


Figure 1. Fabrication and morphology of inkjet-printed perovskite NW arrays. a) The fabrication process of printed perovskite NW arrays. b) Schematics of perovskite NW arrays within the AAO nanopores. c) Cross-sectional SEM observation of the perovskite NWs within the AAO nanostructure.

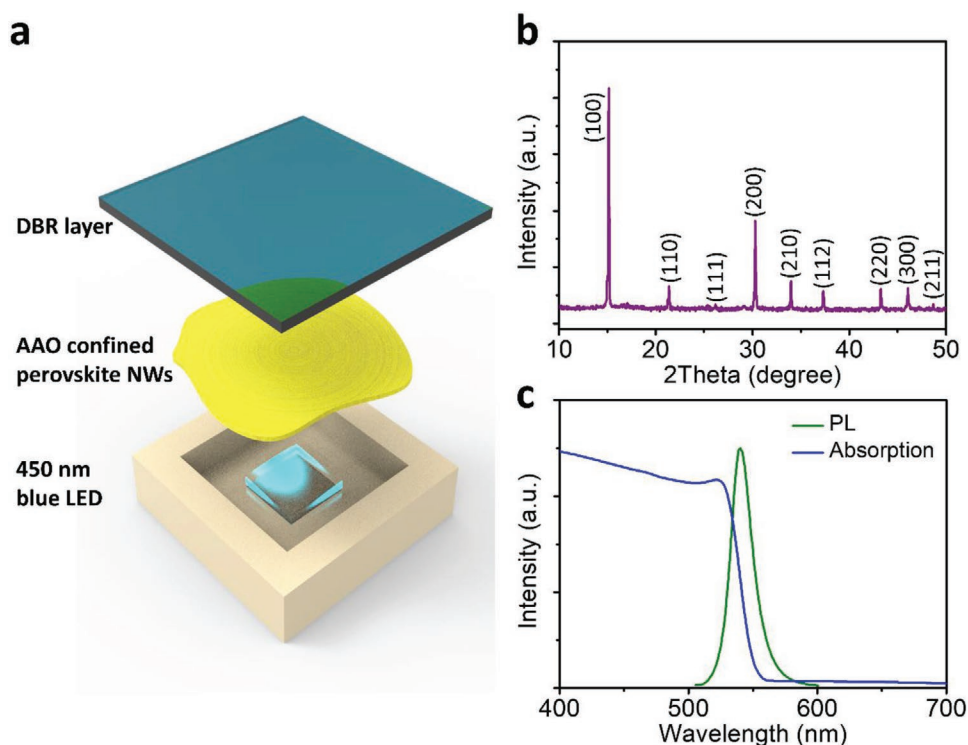


Figure 2. Device design and optical properties of perovskite NW arrays. a) Schematics of the through-plane optical device setup using GaN LED, perovskite NW/AAO, and DBR layer. b) XRD measurement of $\text{CH}_3\text{NH}_3\text{PbBr}_3$ perovskite NW arrays. c) PL and UV-vis absorption spectra of perovskite NW/AAO samples.

Figure 2c demonstrates the PL and UV-vis spectra of $\text{CH}_3\text{NH}_3\text{PbBr}_3$ NW/AAO sample. One can see that the perovskite NW arrays emitted bright green light with an FWHM of 22 nm and peak wavelength around 540 nm, which matches to the UV-vis absorption edge of the perovskite NWs. Moreover, the strong absorption in the blue wavelength region implies that the perovskite NWs can be efficiently pumped by the 450 nm LED. To understand the impact of AAO in the optical device, we measured the transmittance spectrum of a pure AAO substrate as shown in Figure S1 (Supporting Information). The AAO features transmittance of 76% at 540 nm green wavelength, implying that the AAO did not strongly hinder the PL emission from perovskite NWs. For the 450 nm blue light, the AAO is transparent with a slightly decreased transmittance ($\approx 70\%$). However, with a DBR layer to reflect blue light, the optical excitation on perovskite NWs can be maximized. For UV wavelength region below 400 nm, the transmittance dramatically drops due to the AAO absorption, indicating that the blue LED is more suitable for optically pumping $\text{CH}_3\text{NH}_3\text{PbBr}_3$ NW/AAO samples than other UV light sources.

The perovskite NW array can be readily lit up by the blue LED (Figure 3a). Figure 3b demonstrates the emission spectra of the perovskite NW through-plane device when the LED drive current changes from 10 mA to 100 mA. The device reveals stable emission with a peak wavelength of 540 nm under different current levels. Moreover, the current versus PL intensity curve (Figure 3c) shows linear relationship, indicating that as the PL of perovskite NW device is not saturated at high intensity light illumination, the blue LED can efficiently excite the NW arrays. To

examine the device performance, we measured the PL quantum yield (QY) of perovskite NW/AAO film using a spectrometer equipped with 450 W xenon lamp. The obtained QY was 30%, which is not as high as pure perovskite NWs because the AAO can absorb parts of light especially in the UV wavelength region, thus reducing the QY value. We can further determine the PL EQE of device by multiplying the obtained QY with the EQE of blue LEDs. The current dependent PL EQE displays a maximum EQE of 21.6% at 30 mA (Figure 3d), which is higher than that of state-of-the-art electroluminescence (EL) perovskite devices.^[32–34] Generally, the EL device uses perovskite as the active emitter for lighting while PL device takes perovskite as phosphor light converter for white LED and display applications. Since ionic perovskites are relatively unstable under continuous electroluminescence excitation,^[35] up to date, perovskite PL converter devices normally possess longer lifetime and higher efficiency than EL devices.^[35–38] The ambient stability of perovskite NW arrays was also tested. As shown in Figure S2 (Supporting Information), the PL emission from perovskite NWs within the AAO still remained 96% of original intensity after 4 days, while PL from printed perovskite on flat Al_2O_3 substrate decayed to 71% under the same circumstance. Therefore, we can conclude that the AAO provides a remarkable encapsulation effect for the perovskite NW arrays inside the nanopores.

Because the printed perovskite NW arrays exhibit ordered arrangement in the AAO, the polarized PL emission can be theoretically achieved from the perovskite NWs. The in-plane device setup for the polarization measurement is shown in Figure 4a. The 450 nm excitation light was incident along the

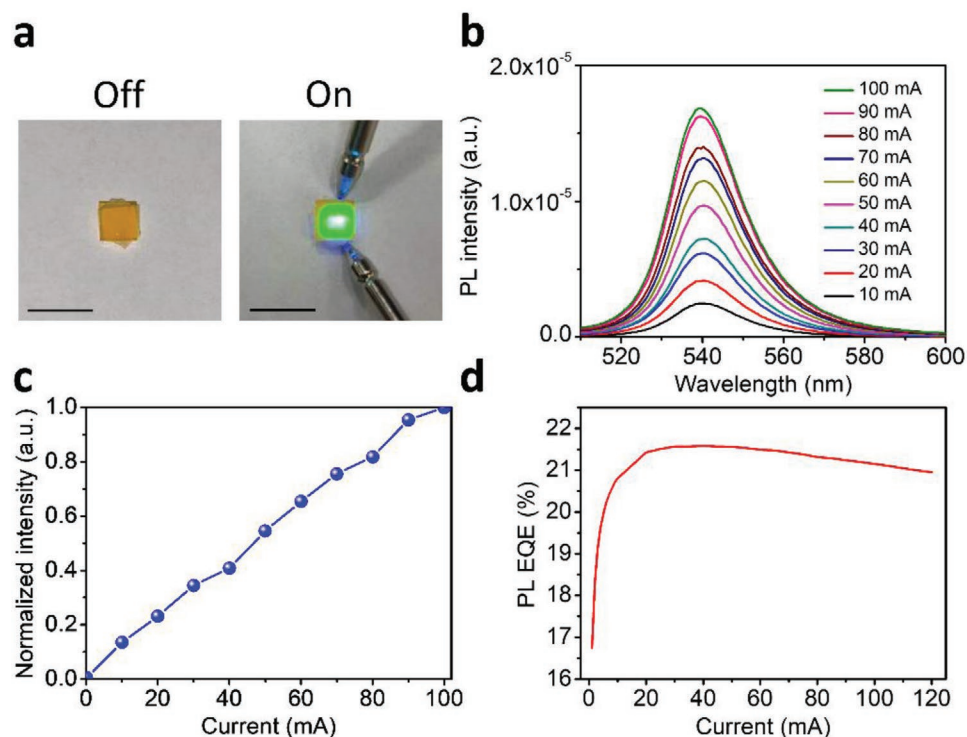


Figure 3. PL characterization of the perovskite NW/AAO device pumped by GaN blue LED. a) Optical images showing the device turned on and off. The length of the scale bars is 5 mm. b) PL spectra of perovskite NW arrays as the LED drive current changes from 10 to 100 mA. c) PL intensity of perovskite NW arrays as a function of drive current. d) PL EQE of the perovskite NW/AAO device as a function of drive current.

in-plane orientation of the sample (perpendicular to the NW long axis) and a photodetector was employed to detect the PL intensity as the polarizer rotated. Along with rotation of the polarizer from 0° to 180° , the blue LED light source was kept identical, but the obtained PL signal from perovskite NW arrays demonstrates a cycle of sine fluctuation as shown in Figure 4b, indicating the high polarization effect.^[39] The DOLP can be determined by Equation (1)^[23]

$$P = (I_{\max} - I_{\min}) / (I_{\max} + I_{\min}) \quad (1)$$

where $I_{\max}$ and $I_{\min}$ are the maximum and minimum values of obtained PL intensity in the polarization measurement.

The calculated DOLP is 0.84, confirming the high polarization anisotropy of perovskite NW array films. The polarization of 1D perovskite NWs can be attributed to their anisotropic shape and dielectric difference between NWs and the surrounding environment.^[20] When the light wave propagates in the NW, the electric field perpendicular to the NW long axis ($E_{\perp}$) is attenuated due to mismatched dielectric constants, while electric field parallel to the NW long axis ($E_{\parallel}$) remains the same.^[40] The intensity ratio of $E_{\perp}$ and $E_{\parallel}$ can be described by^[2]

$$\frac{E_{\perp}}{E_{\parallel}} = \frac{2\varepsilon_0}{\varepsilon + \varepsilon_0} \quad (2)$$

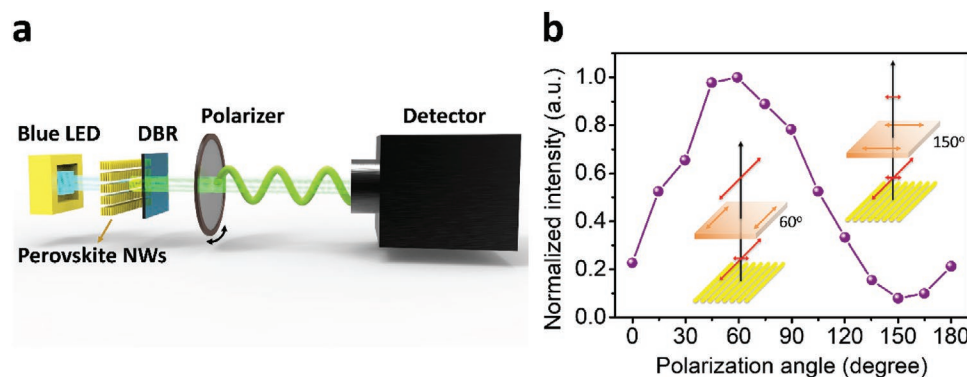


Figure 4. Polarization measurement of perovskite NW arrays. a) Schematic of the polarization experimental setup. b) Polarized emission property of perovskite NW arrays. The inset illustrates the polarization measurement when the angle of polarizer located at 60° and 150° .

where ϵ and ϵ_0 represent the dielectric constants of NW and the environment, respectively. Therefore, the field distribution in perovskite NWs is essentially nonuniform, which leads to the anisotropic polarization property.

Table 1 summarizes the reports using perovskite for polarized emission. Although high DOLP ≈ 0.7 was reported based on an individual perovskite NW,^[20] the DOLP for a cluster of unordered NWs are still lower than 0.5,^[21,40] demonstrating the importance of structural alignment for polarization applications. Gao et al. have reported an orientationally grown of CsPbBr₃ NW arrays on muscovite mica using chemical vapor deposition (CVD), which possesses the capability to achieve high DOLP of 0.78.^[41] However, compared with simple and rapid inkjet printing, the CVD method is costly, time-consuming, and requires careful control of deposition parameters. It is worth noting that perovskite quantum dots (QDs) can also exhibit anisotropic polarization in some cases. Because of cubic lattice structure, perovskite QD itself does not feature polarization property. However, many previous researches have proven that the ionic movement (MA⁺, Pb²⁺, X⁻) does exist in the perovskite, which gives rise to the change of electric field distribution.^[23,25] It will break the space field symmetry and yield the dipole moment, thus resulting in the anisotropic polarization in perovskite. Additionally, when large amounts of QDs gather, they tend to attract each other and self-assemble to ordered agglomerates in a large area, which can enable polarized emission without further alignment processes.^[23] As a result, several groups have observed the polarization property of perovskite QDs with DOLP values from 0.28 to 0.54.^[2,23,39,42]

In addition to 3D cubic perovskite we just mentioned, 2D perovskite is another perovskite family that possesses good polarization property because of its asymmetric layered crystal structure. For example, Ghoshal et al. demonstrated a high DOLP of 0.73 from an individual (BA)₂PbI₄ NW,^[43] and Ma et al. reported polarized PL based on a group of chiral (MBA)₂PbI₄ microplates with a DOLP of 0.176.^[44] Compared with perovskite-based devices in the literature, the printed perovskite NW arrays in this study present the highest DOLP of 0.84, suggesting its potential for polarized light application. Using highly polarized backlight in LCD panels, the lighting

efficiency can be significantly increased. For example, Lu et al. have reported the polarized emission from MAPbBr₃ QDs/polyvinylidene fluoride (PVDF) composite film (DOLP = 0.33), which can reduce the optical loss of polarizer conversion from 50% to 35% in the LCD.^[2] Consequently, using the perovskite NW arrays as polarized light source, the optical loss can be further reduced, thus enhancing the efficiency greatly.

3. Conclusion

In summary, by incorporating inkjet printing and the AAO confinement strategy, we successfully fabricated highly ordered perovskite NW array films. With blue LED, perovskite NW/AAO film, and DBR layer, the optical device using through-plane design presents high PL EQE of 21.6%, confirming its superior lighting capability. Due to very anisotropic 1D NW shape and good alignment of perovskite NW array films, the extremely polarized emission was observed on the in-plane optical device with the DOLP up to 0.84, which is much higher than any other perovskite lighting devices in the literature. With unique anisotropic property and excellent lighting performance in terms of efficiency and polarization, our results would be of great interest to optoelectronics community, encouraging the development of highly polarized perovskite devices for display and polarized light application.

4. Experimental Section

Inkjet-Printed CH₃NH₃PbBr₃ NW Arrays: First, 1.12 g of CH₃NH₃Br and 3.67 g of PbBr₂ powders were dissolved in 20 mL dimethylformamide at 90 °C to prepare perovskite inks. Then, the perovskite inks were injected into printer cartridge through a 0.25 μ m filter. An inkjet printer (Fujifilm, DMP-2831) was employed to print the perovskite inks on the top of 0.02 μ m pore size AAO substrate with a bottom vacuum pumping to guide the inks into AAO nanopores. The sample was put on a 90 °C hotplate for 2 h to remove dimethylformamide solvent and the perovskite NW arrays were recrystallized in the AAO nanopores.

Characterizations of Inkjet-Printed CH₃NH₃PbBr₃ NW Arrays: The morphology of perovskite NWs was studied by using a Magellan extreme high-resolution SEM with voltage setting of 5 kV. The XRD measurement of perovskite sample was carried out using a Bruker D2 PHASER diffractometer. The absorption spectrum of perovskite NWs and transmission spectrum of AAO were examined by Shimadzu UV 3600 spectrophotometer. PL spectrum of perovskite NW arrays was obtained using an Edinburgh Instruments Spectrofluorometer FS5 equipped with a 150 W xenon lamp. The QY measurement was undertaken using an SPEX fluorolog-3 spectrometer (Jobin Yvon Instrument) combined with a 450 W xenon lamp and an R928 photodetector (Hamamatsu Photonics).

Characterization of Perovskite NW Device: The PL EQE of optical device was determined by multiplying obtained QY of perovskite NW/AAO sample with the EQE of blue LED. The EQE measurement of blue LED was performed using a spectrometer CAS 140CT (Instrument System GmbH, Germany) equipped with a 50 cm integrating sphere. The polarization experimental setup is shown in Figure 4a. The 450 nm GaN LED (Epistar Corporation, Taiwan, chip size: 500 μ m $\times$ 1000 μ m) and a blue-cut DBR film was employed to excite the perovskite NW arrays. The PL emission from perovskite NW arrays transmitted a polarizer and detected by a spectrometer (Hitachi U4100, Japan). By rotating the polarizer, the PL intensity as a function of angle can be obtained (Figure S3, Supporting Information). The maximum and minimum intensities were recorded at 60° and 150°, respectively.

Table 1. An overview of perovskite materials for polarized emission.

Materials	Emission peak	Polarization (DOLP)	Ref.
MAPbBr ₃ NW arrays/AAO	540 nm	0.84	This work
Individual (BA) ₂ PbI ₄ NW	518 nm	0.73	[43]
Individual MAPbI ₃ NW	760 nm	0.7	[20]
CsPbBr ₃ NW Arrays/mica	515 nm	0.78	[41]
CsPbBr ₃ NWs/PLMA polymer	517 nm	0.44	[40]
CsPbBr ₃ NWs/PU fiber	490 nm	0.3	[21]
MAPbBr ₃ QD/PVA fiber	$\approx$ 525 nm	0.53	[42]
CsPbI ₃ QD film	693 nm	0.4	[23]
MAPbBr ₃ QD/PVDF	528 nm	0.33	[2]
MAPbI ₃ QD solution	711 nm	0.28	[39]
(MBA) ₂ PbI ₄ microplates	$\approx$ 510 nm	0.176	[44]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was financially supported by the Ministry of Science and Technology, Taiwan (MOST) (107-2221-E-009-113-MY3), Australian Research Council (ARC) (DP190103316), and the startup funding of City University of Hong Kong. The authors would like to thank Everlight Electronics for helpful technical support with the polarized optical devices.

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

C.H.L. and C.Y.K. contributed equally to this work. C.H.L. fabricated the perovskite NW arrays and studied its morphology and optical properties. C.Y.K., T.Z.W., C.L.T., and C.W.S. fabricated the optical device based on perovskite NW arrays, measured the device performance, and studied the polarized emission. C.H.L., C.Y.K., X.G., P.T.L., T.W., C.H.H., J.H.H., and H.C.K. analyzed the results, prepared the figures, and wrote the manuscript.

Keywords

anisotropic applications, anisotropic polarization, light emission, perovskite nanowire array films

Received: November 7, 2019

Revised: December 31, 2019

Published online: February 19, 2020

- [1] A. K. Srivastava, W. L. Zhang, J. Schneider, A. L. Rogach, V. G. Chigrinov, H. S. Kwok, *Adv. Mater.* **2017**, 29, 1701091.
- [2] W. G. Lu, X. G. Wu, S. Huang, L. Wang, Q. C. Zhou, B. S. Zou, H. Z. Zhong, Y. T. Wang, *Adv. Opt. Mater.* **2017**, 5, 1700594.
- [3] M. X. Ren, W. Wu, W. Cai, B. Pi, X. Z. Zhang, J. J. Xu, *Light: Sci. Appl.* **2017**, 6, e16254.
- [4] Y. Q. Wang, C. Yang, Y. G. Wang, N. Chi, *Opt. Lett.* **2014**, 39, 1823.
- [5] H. T. Yuan, X. G. Liu, F. Afshinmanesh, W. Li, G. Xu, J. Sun, B. Lian, A. G. Curto, G. J. Ye, Y. Hikita, Z. X. Shen, S. C. Zhang, X. H. Chen, M. Brongersma, H. Y. Hwang, Y. Cui, *Nat. Nanotechnol.* **2015**, 10, 707.
- [6] J. G. Feng, X. X. Yan, Y. Liu, H. F. Gao, Y. C. Wu, B. Su, L. Jiang, *Adv. Mater.* **2017**, 29, 1605993.
- [7] Y. F. Wang, J. F. Li, S. C. Zhang, K. Y. Su, Y. R. Zhou, K. Y. Liao, S. W. Du, H. Yan, S. L. Zhu, *Nat. Photonics* **2019**, 13, 346.
- [8] E. Knill, R. Laflamme, G. J. Milburn, *Nature* **2001**, 409, 46.
- [9] A. Sitt, A. Salant, G. Menagen, U. Banin, *Nano Lett.* **2011**, 11, 2054.
- [10] I. Hadar, J. P. Philbin, Y. E. Panfil, S. Neyshadt, I. Lieberman, H. Eshet, S. Lazar, E. Rabani, U. Banin, *Nano Lett.* **2017**, 17, 2524.
- [11] P. D. Cunningham, J. B. Souza, I. Fedin, C. X. She, B. Lee, D. V. Talapin, *ACS Nano* **2016**, 10, 5769.
- [12] T.-C. Wei, S. Mokkapat, T.-Y. Li, C.-H. Lin, G.-R. Lin, C. Jagadish, J.-H. He, *Adv. Funct. Mater.* **2018**, 28, 1707175.
- [13] C. H. Lin, A. Verma, C. Y. Kang, Y. M. Pai, T. Y. Chen, J. J. Yang, C. W. Sher, Y. Z. Yang, P. T. Lee, C. C. Lin, Y. C. Wu, S. K. Sharma, T. Wu, S. R. Chung, H. C. Kuo, *Photonics Res.* **2019**, 7, 579.
- [14] X. Guan, Z. Wang, M. K. Hota, H. N. Alshareef, T. Wu, *Adv. Electron. Mater.* **2019**, 5, 1800538.
- [15] B. Cheng, T.-Y. Li, P. Maity, P.-C. Wei, D. Nordlund, K.-T. Ho, D.-H. Lien, C.-H. Lin, R.-Z. Liang, X. Miao, I. A. Ajia, J. Yin, D. Sokaras, A. Javey, I. S. Roqan, O. F. Mohammed, J.-H. He, *Commun. Phys.* **2018**, 1, 80.
- [16] G. Long, C. Jiang, R. Sabatini, Z. Yang, M. Wei, L. N. Quan, Q. Liang, A. Rasmita, M. Askerka, G. Walters, X. Gong, J. Xing, X. Wen, R. Quintero-Bermudez, H. Yuan, G. Xing, X. R. Wang, D. Song, O. Voznyy, M. Zhang, S. Hoogland, W. Gao, Q. Xiong, E. H. Sargent, *Nat. Photonics* **2018**, 12, 528.
- [17] S. Niu, G. Joe, H. Zhao, Y. Zhou, T. Orvis, H. Huyen, J. Salman, K. Mahalingam, B. Urwin, J. Wu, Y. Liu, T. E. Tiwald, S. B. Cronin, B. M. Howe, M. Mecklenburg, R. Haiges, D. J. Singh, H. Wang, M. A. Kats, J. Ravichandran, *Nat. Photonics* **2018**, 12, 392.
- [18] N. Zhou, Y. Bekenstein, C. N. Eisler, D. Zhang, A. M. Schwartzberg, P. Yang, A. P. Alivisatos, J. A. Lewis, *Sci. Adv.* **2019**, 5, eaav8141.
- [19] H. T. Yi, S. Rangan, B. Tang, C. D. Frisbie, R. A. Bartynski, Y. N. Gartstein, V. Podzorov, *Mater. Today* **2019**, 28, 31.
- [20] D. Tauber, A. Dobrovolsky, R. Camacho, I. G. Scheblykin, *Nano Lett.* **2016**, 16, 5087.
- [21] T. Guner, G. Topcu, U. Savaci, A. Genc, S. Turan, E. Sari, M. M. Demir, *Nanotechnology* **2018**, 29, 135202.
- [22] C. Y. Chen, J. H. Huang, J. H. Song, Y. S. Zhou, L. Lin, P. C. Huang, Y. Zhang, C. P. Liu, J. H. He, Z. L. Wang, *ACS Nano* **2011**, 5, 6707.
- [23] D. Wang, D. Wu, D. Dong, W. Chen, J. J. Hao, J. Qin, B. Xu, K. Wang, X. W. Sun, *Nanoscale* **2016**, 8, 11565.
- [24] D. Periyangounder, T.-C. Wei, T.-Y. Li, C.-H. Lin, T. P. Gonçalves, H.-C. Fu, D.-S. Tsai, J.-J. Ke, H.-W. Kuo, K.-W. Huang, N. Lu, X. Fang, J.-H. He, *Adv. Mater.* **2019**, 1904634.
- [25] J. You, L. Meng, T.-B. Song, T.-F. Guo, Y. Yang, W.-H. Chang, Z. Hong, H. Chen, H. Zhou, Q. Chen, Y. Liu, N. D. Marco, Y. Yang, *Nat. Nanotechnol.* **2016**, 11, 75.
- [26] A. Waleed, M. M. Tavakoli, L. Gu, S. Hussain, D. Zhang, S. Poddar, Z. Wang, R. Zhang, Z. Fan, *Nano Lett.* **2017**, 17, 4951.
- [27] C. H. Lin, B. Cheng, T. Y. Li, J. R. D. Retamal, T. C. Wei, H. C. Fu, X. Fang, J. H. He, *ACS Nano* **2019**, 13, 1168.
- [28] M. X. Ren, M. Chen, W. Wu, L. H. Zhang, J. K. Liu, B. Pi, X. Z. Zhang, Q. Q. Li, S. S. Fan, J. J. Xu, *Nano Lett.* **2015**, 15, 2951.
- [29] L. X. Chen, Y. Y. Rong, M. X. Ren, W. Wu, M. Y. Qin, C. D. Pan, Q. Ma, S. K. Liu, B. T. Wu, E. Wu, J. J. Xu, H. P. Zeng, *J. Phys. Chem. C* **2018**, 122, 15666.
- [30] C.-H. Lin, H.-C. Fu, B. Cheng, M.-L. Tsai, W. Luo, L. Zhou, S.-H. Jang, L. Hu, J.-H. He, *npj 2D Mater. Appl.* **2018**, 2, 23.
- [31] T. C. Wei, H. P. Wang, T. Y. Li, C. H. Lin, Y. H. Hsieh, Y. H. Chu, J. H. He, *Adv. Mater.* **2017**, 29, 1701789.
- [32] K. B. Lin, J. Xing, L. N. Quan, F. P. G. de Arquer, X. W. Gong, J. X. Lu, L. Q. Xie, W. J. Zhao, D. Zhang, C. Z. Yan, W. Q. Li, X. Y. Liu, Y. Lu, J. Kirman, E. H. Sargent, Q. H. Xiong, Z. H. Wei, *Nature* **2018**, 562, 245.
- [33] Y. Cao, N. Wang, H. Tian, J. S. Guo, Y. Wei, H. Chen, Y. Miao, W. Zou, K. Pan, Y. He, H. Cao, Y. Ke, M. Xu, Y. Wang, M. Yang, K. Du, Z. Fu, D. Kong, D. Dai, Y. Jin, G. Li, H. Li, Q. Peng, J. Wang, W. Huang, *Nature* **2018**, 562, 249.
- [34] B. Zhao, S. Bai, V. Kim, R. Lamboll, R. Shivanna, F. Auras, J. M. Richter, L. Yang, L. Dai, M. Alsari, X.-J. She, L. Liang, J. Zhang, S. Lilliu, P. Gao, H. J. Snaith, J. Wang, N. C. Greenham, R. H. Friend, D. Di, *Nat. Photonics* **2018**, 12, 783.

- [35] X. F. Zhao, J. D. A. Ng, R. H. Friend, Z. K. Tan, *ACS Photonics* **2018**, 5, 3866.
- [36] C.-Y. Kang, C.-H. Lin, C.-H. Lin, T.-Y. Li, S.-W. H. Chen, C.-L. Tsai, C.-W. Sher, T.-Z. Wu, P.-T. Lee, X. Xu, M. Zhang, C.-H. Ho, J.-H. He, H.-C. Kuo, *Adv. Sci.* **2019**, 6, 1902230.
- [37] Y. Wei, Z. Y. Cheng, J. Lin, *Chem. Soc. Rev.* **2019**, 48, 405.
- [38] C. H. Lin, C. Y. Kang, A. Verma, T. Wu, Y. M. Pai, T. Y. Chen, C. L. Tsai, Y. Z. Yang, S. K. Sharma, C. W. Sher, Z. Chen, P. T. Lee, S. R. Chung, H. C. Kuo, *Nanomaterials* **2019**, 9, 1314.
- [39] Z. F. Shi, Y. Li, S. Li, H. F. Ji, L. Z. Lei, D. Wu, T. T. Xu, J. M. Xu, Y. T. Tian, X. J. Li, *J. Mater. Chem. C* **2017**, 5, 8699.
- [40] S. N. Raja, Y. Bekenstein, M. A. Koc, S. Fischer, D. Zhang, L. Lin, R. O. Ritchie, P. Yang, A. P. Alivisatos, *ACS Appl. Mater. Interfaces* **2016**, 8, 35523.
- [41] Y. Gao, L. Y. Zhao, Q. Y. Shang, Y. G. Zhong, Z. Liu, J. Chen, Z. P. Zhang, J. Shi, W. N. Du, Y. F. Zhang, S. L. Chen, P. Gao, X. F. Liu, X. N. Wang, Q. Zhang, *Adv. Mater.* **2018**, 30, 1801805.
- [42] L. H. Meng, C. G. Yang, J. J. Meng, Y. Z. Wang, Y. Ge, Z. Q. Shao, G. F. Zhang, A. L. Rogach, H. Z. Zhong, *Nano Res.* **2019**, 12, 1411.
- [43] D. Ghoshal, T. Wang, H. Z. Tsai, S. W. Chang, M. Crommie, N. Koratkar, S. F. Shi, *Adv. Opt. Mater.* **2019**, 7, 1900039.
- [44] J. Q. Ma, C. Fang, C. Chen, L. Jin, J. Q. Wang, S. Wang, J. Tang, D. H. Li, *ACS Nano* **2019**, 13, 3659.

基于微孔硅阵列的像素化 γ -CuI 闪烁转换屏的研制及性能表征

孙寿强, 顾 牡*, 陈婷婷, 张娟楠, 刘小林, 刘 波, 黄世明, 倪 晨

同济大学物理科学与工程学院, 上海市特殊人工微结构材料与技术重点实验室, 上海 200092

摘 要 以微孔硅阵列模板, 高纯 CuI 粉末为原料, 采用压力注入法, 成功制备了具有单分散微柱结构的像素化 CuI 闪烁转换屏。扫描电子显微镜(SEM)与 X 射线衍射(XRD)的测试结果表明, 所制备的转换屏中 CuI 微柱连续、致密, 微柱柱径约为 $2.5\ \mu\text{m}$ 、间隔约为 $1.5\ \mu\text{m}$ 、柱长约为 $80\ \mu\text{m}$, 并具有良好的 γ 相晶体结构。在 X 射线激发下, 所制备的像素化 γ -CuI 闪烁转换屏具有峰值波长位于 $680\ \text{nm}$ 附近的红光慢发射带; 掺碘后, 该发射带被较大幅度的抑制, 同时出现了峰值波长位于 $432\ \text{nm}$ 的快发光峰; 当碘掺杂含量达到 $10\ \text{Wt}\%$ 时, 峰值波长位于 $680\ \text{nm}$ 附近的红光慢发射带被完全抑制, 只存在峰值波长位于 $432\ \text{nm}$ 的快发光峰。采用刃边法测量了所制备的像素化 γ -CuI 闪烁屏的空间分辨率, 结果显示其分辨率可达 $38\ \text{lp}\cdot\text{mm}^{-1}$, 表明该闪烁屏除拥有超快时间响应特性外, 兼具很高的空间分辨本领, 在 X 射线成像方面具有独特的应用价值。

关键词 像素化 γ -CuI 闪烁转换屏; 微柱结构; 闪烁性能; 空间分辨率

中图分类号: O482.3 文献标识码: A DOI: 10.3964/j.issn.1000-0593(2017)05-1566-05

引 言

X 射线成像在惯性约束聚变、医学诊断、无损检测、国防安检和天体物理等诸多领域有着十分重要的应用^[1]。闪烁转换屏可将 X 射线图像转换成可见光图像, 因而是数字化 X 射线成像系统的核心部件, 其性能是影响数字 X 射线成像质量的关键。为了提高 X 射线闪烁转换屏的空间分辨率, 人们常通过限制闪烁屏的厚度以降低闪烁光的侧向传播^[2], 然而较薄的厚度又很难满足闪烁转换屏对 X 射线探测效率的需要, 因此普通的闪烁转换屏难以同时满足高空间分辨率和高 X 射线探测效率的要求。为了解决这一难题, 人们开展了具有微柱结构像素化闪烁转换屏的研究^[3], 垂直于屏面的微柱可引导闪烁光沿微柱柱向传播, 从而可限制闪烁光的侧向传播。微柱结构像素化闪烁转换屏的制备主要有激光切割法^[4]、基底图案法^[5]、调控蒸发法^[6]、微孔氧化硅模板填充法^[7]等。鉴于微孔氧化硅模板不仅可利用单分散微柱边界的全反射, 有效地引导闪烁光沿柱向传播, 而且对于那些透射到孔壁内的闪烁光, 也能通过硅壁将其吸收, 从而可有效抑制闪烁光的侧向传播^[7], 因此采用该模板填充法制备像素化转换屏被认为是一条比较理想的技术路线。关于这方面的研究, 目前主要集中在基于大孔硅阵列模板 CsI:Tl 闪烁转换

屏的研制上, 但从已有的结果来看, 微柱柱径基本在几十微米量级, 微柱内尚存有气泡等现象^[7], 因此制备工艺有待进一步优化, 以获得微柱柱径更小、填充更为致密的像素化闪烁转换屏。

γ -CuI 是一种宽禁带 p 型半导体 (禁带宽度 $E_g = 3.1\ \text{eV}$), 具有较高的激子结合能和快离子导电等特性。随着该材料发射峰位于 $410\sim 430\ \text{nm}$ 、发光衰减时间仅为 $90\ \text{ps}$ 的超快闪烁成份的发现, γ -CuI 已成为已知最快的无机闪烁材料之一^[8]。

为了获得超高空间和时间分辨的闪烁转换屏, 采用压力注入法, 通过制备工艺的摸索, 制备出了基于微孔硅阵列的像素化 CuI 闪烁转换屏; 并对该闪烁转换屏的形貌、结晶性、X 射线激发发射性能进行了表征分析; 通过碘掺杂优化了 γ -CuI 转换屏的闪烁性能; 采用刃边法测试了所制备闪烁转换屏的空间分辨率。

1 实验部分

1.1 样品制备

采用压力注入法制备像素化 CuI 闪烁转换屏。自行搭建的压力注入装置原理图可参见文献^[9], 其制备流程如图 1 所示。首先, 采用图案引导电化学刻蚀的方法^[10], 制备周期

收稿日期: 2015-10-09, 修订日期: 2016-03-16

基金项目: 国家自然科学基金项目(11475128, 11375129, 11475127)和科学技术部国家重大科学仪器设备开发专项(2011YQ13001902)资助

作者简介: 孙寿强, 1991 年生, 同济大学物理科学与工程学院硕士研究生 e-mail: ssqaxq1991@163.com

* 通讯联系人 e-mail: mgu@tongji.edu.cn

排列、均匀分布的微孔硅阵列模板：为了提高微柱界面对闪烁光的全反射性能，将刻蚀获得的微孔硅模板进行高温氧化（1100 °C，5 h），在硅孔内表面形成一层均匀致密的二氧化硅薄层，得到微孔氧化硅阵列模板[图 1(a)]。然后，将模板固定于压力注入装置的底部，取 10 g 纯度高于 99.99% 的 CuI 粉末原料均匀平铺于模板上表面[图 1(b)]，接着，将压力装置腔内空气抽到真空度低于 10^{-2} Torr，加热原料至 610 °C，并保持约 10 min 使得 CuI 粉末完全熔融，这时施加约 2 MPa 氩气作用于熔体上表面，使得熔融 CuI 在外界压力作用下连续致密地流入到模板的微孔中[图 1(c)]，持续一段时间待填充完毕后，关闭加热和停止加压，待样品自然冷却至室温后取出；通过机械抛光除去模板表面残余 CuI 原料，得到像素化 CuI 闪烁转换屏样品[图 1(d)]。此外，在 CuI 原料中均匀掺入一定量的碘单质，通过同样的实验过程，可获得碘掺杂的闪烁转换屏样品。

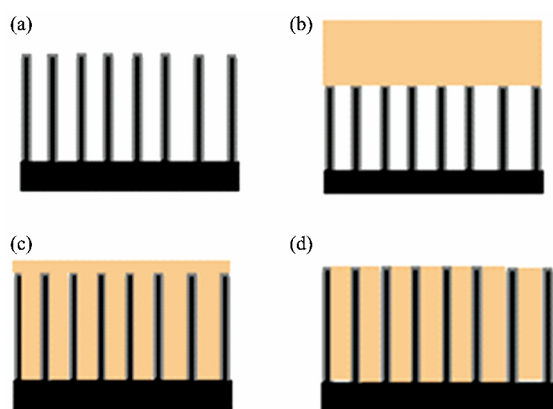


图 1 像素化 CuI 闪烁转换屏的制备流程图

(a): 微孔氧化硅阵列模板；(b): 压力注入前；(c): 压力注入后；(d): 像素化 CuI 闪烁转换屏

Fig. 1 Schematic illustrations of the step-by-step fabrication processes

(a): Oxidized silicon micropore template; (b): Prior to injection; (c): After injection; (d): The pixelated CuI scintillation screen

1.2 表征测试

采用荷兰 Philips XL-30 扫描电子显微镜 (scanning electron microscope, SEM) 分别观察了微孔硅阵列模板和闪烁转换屏样品的表面、断面形貌；采用 DX-2700 型 X 射线衍射仪 (X-ray diffractometer, XRD) 测试了转换屏的结晶性能；采用自行研制的 X 射线激发发射谱仪测试了转换屏的 X 射线激发发射谱 (X-ray excited luminescence, XEL)，其中 X 射线管的工作电压为 80 kV，工作电流为 4 mA。

微柱结构像素化 γ -CuI 闪烁转换屏的空间分辨率表征在上海光源 BL13W1 光束线站上进行，测量中采用了刃边法^[11]和由 Flash 4.0 型 CCD 相机和显微系统构成的法国 Optique Peter 探测器，X 射线的能量为 20 keV，曝光时间为 10 s。所有测试均在室温下进行。

2 结果与讨论

图 2(a) 和 (b) 是高温氧化后具有 SiO₂ 反射层结构的微孔硅阵列模板的表面、断面 SEM 形貌图，可以看到，模板由周期性均匀分布的微孔组成，微孔排列周期（孔间距）为 4 μm 、孔径约为 2.5 μm 、壁厚约为 1.5 μm ，其中孔壁表面 SiO₂ 层厚度约为 500 nm，如图 2(c)，具有 SiO₂ 氧化层的微孔硅阵列，可同时利用界面的全反射和孔壁中间的硅层对可见光的吸收，抑制闪烁光的横向传播如图 2(d)。

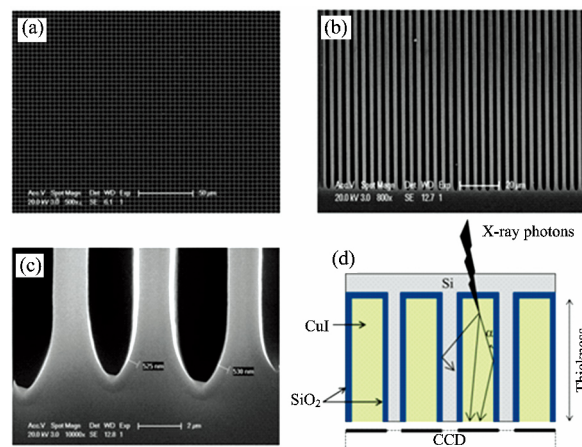


图 2 微孔氧化硅阵列模板的 (a) 表面形貌图，(b) 断面形貌图和 (c) 局部放大图，(d) 为闪烁转换屏探测原理图

Fig. 2 (a) Surface morphology, (b) cross-sectional view and (c) partial enlarged view of the oxidized silicon micropore template, (d) schematic of the detection principle

以此微孔硅阵列为模板制备的像素化 CuI 闪烁转换屏的表面、断面形貌如图 3，由图可见，所有的硅微通道均被 CuI 完全填充，形成均匀、致密、连续的单分散微柱结构，微柱径约为 2.5 μm 、间隔约为 1.5 μm 、厚度约为 80 μm 的像素化 CuI 闪烁转换屏。

图 4 为所制备闪烁转换屏样品的 X 射线衍射谱，由图可见，样品的所有衍射峰均与标准卡片 Na₀₆₋₀₂₄₆ 完全吻合，表明闪烁转换屏中的 CuI 属于 γ 相闪锌矿晶体结构。另外，图谱中没有出现衍射峰宽化及其他任何杂峰，表明所制备的样品具有良好的结晶性能和晶相纯度。

图 5(a) 是未掺杂样品的 X 射线激发发射谱，该转换屏在 X 射线的激发下只存在峰值波长位于 680 nm 附近的红光发射带，该发射带被认为源于 γ -CuI 中的碘空位^[12]，其发光衰减时间约为 1.5 μs ，属慢闪烁成份^[13]。为了进一步优化 γ -CuI 转换屏的闪烁性能，尝试在 CuI 原料中分别掺入 2 Wt% 和 10 Wt% 的单质碘，当掺碘含量为 2 Wt% 时， γ -CuI 闪烁转换屏的 X 射线激发发射谱如图 5(b)，可见峰值波长位于 680 nm 附近的红光慢发射带受到了较大程度的抑制，而在 432 nm 处出现了一个相对较强的紫光发射峰，该发光峰被认为源于 γ -CuI 中的施主-受主对 (donor-acceptor pair, DAP) 发光^[12]，其发光衰减时间约为 90 ps，属超快闪烁成分^[8]。

将碘掺杂含量增加至 10 Wt%，发现峰值波长位于 680 nm 附近的红光慢发射带被完全抑制，样品只存在峰值波长位于 432 nm 的快闪烁成份如图 5(c)。

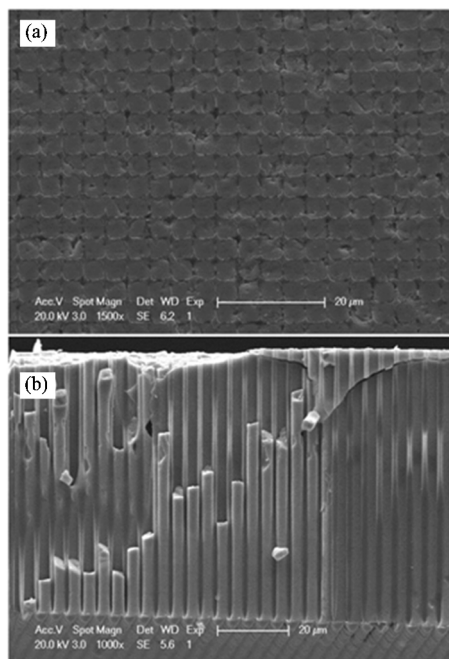


图 3 像素化 γ -CuI 闪烁转换屏 (a) 表面形貌图和 (b) 断面形貌图

Fig 3 (a) Surface morphology and (b) cross-sectional view of the γ -CuI pixelated scintillation screen

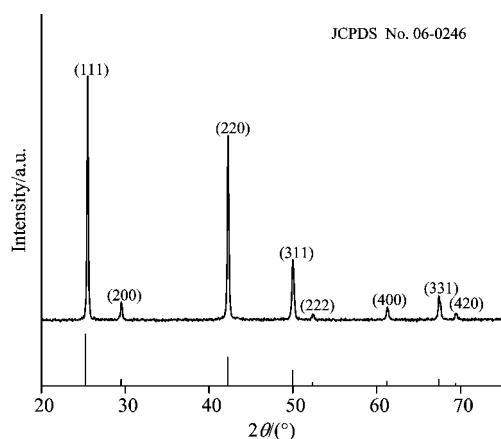


图 4 像素化 γ -CuI 闪烁转换屏的 X 射线衍射谱图

Fig 4 XRD patterns of the γ -CuI pixelated scintillation screen

采用刃边法在上海同步辐射光源装置上测量了所制备转换屏的空间分辨率，实验所得的调制传递函数 (modulation transfer function, MTF) 如图 6，由图可知， γ -CuI 闪烁转换屏的空间分辨率可达 $38 \text{ pl} \cdot \text{mm}^{-1}$ ，与 Badel, Simon 和 Cha 等的结果相比^[3, 7, 14]，由于填充更为致密，微柱的柱径更小，闪烁转换屏 X 射线成像的空间分辨率更高。

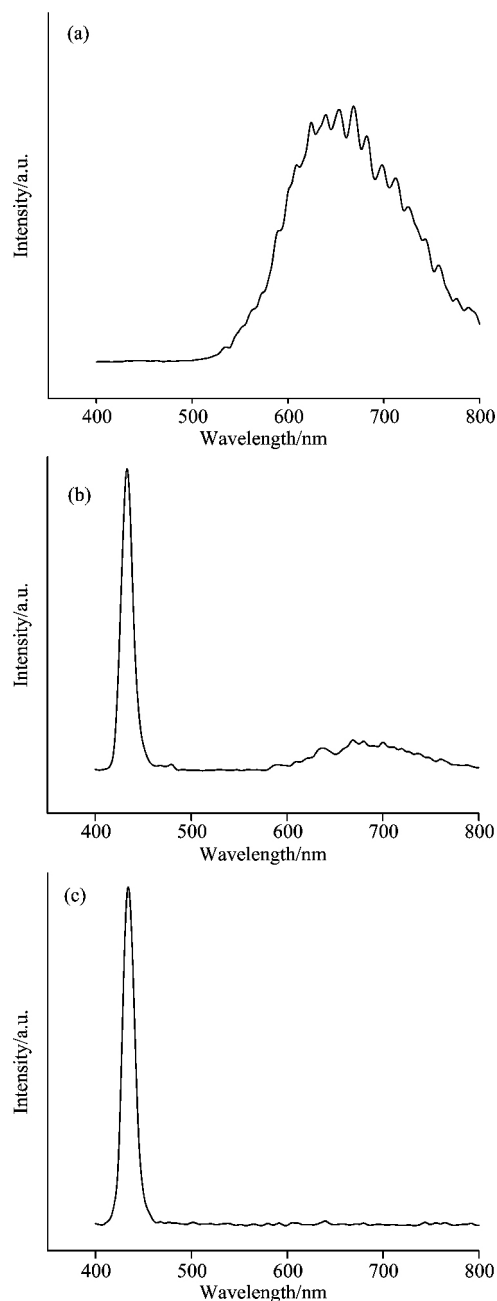


图 5 像素化 γ -CuI 闪烁转换屏的 X 射线激发发射谱

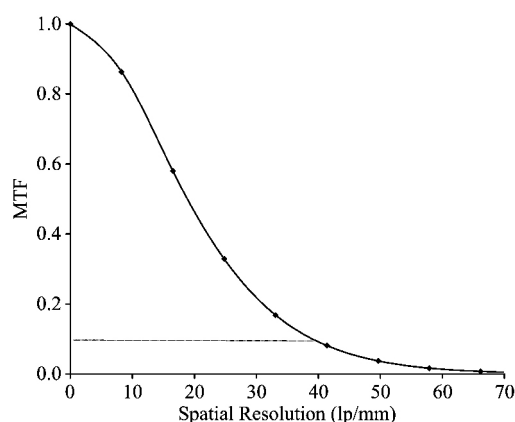
(a): 未掺杂样品; (b): 2 Wt% 碘掺杂样品;
(c): 10 Wt% 碘掺杂样品

Fig 5 XEL of the γ -CuI pixelated scintillation screen

(a): Undoped sample; (b): 2 Wt% iodine doped sample;
(c): 10 Wt% iodine doped sample

3 结 论

以微孔硅阵列模板，采用压力注入法成功地制备出具有单分散微柱结构，柱径约为 $2.5 \mu\text{m}$ 、间隔约为 $1.5 \mu\text{m}$ 、厚度约为 $80 \mu\text{m}$ 的像素化 γ -CuI 闪烁转换屏。扫描电子显微镜 (SEM) 和 X 射线衍射 (XRD) 的结果表明，所制备的闪烁转

图6 像素化 γ -CuI闪烁转换屏的调制传递函数图Fig 6 MTF of the γ -CuI pixelated scintillation screen

换屏不仅具有均匀、连续、致密的CuI微柱形貌,而且还具有良好的 γ 相闪烁晶体结构。X射线激发发射测试结果表明,所制备的 γ -CuI闪烁转换屏只存在峰值波长位于680 nm附近较强的红光慢发射带,但通过碘掺杂能有效地优化转换屏的闪烁性能,当碘掺杂含量为10 Wt%时,680 nm慢发射带被完全抑制,转换屏只呈现较强的峰值波长位于432 nm的超快闪烁成份。刃边法测量显示,该闪烁转换屏的空间分辨率可达 $38 \text{ lp} \cdot \text{mm}^{-1}$ 。综上所述所制备的 γ -CuI闪烁转换屏除拥有超快时间响应特性外,兼具很高的空间分辨本领,在X射线成像方面具有独特的应用价值。

致谢:感谢上海同步辐射光源BL13W1光束线站提供的科学实验平台,以及相关技术人员的大力支持与帮助,在此一并致谢。

References

- [1] Antonuk L, Jee K W, El-Mohri Y, et al. Medical Physics, 2000, 27(2): 289.
- [2] Martin T, Koch A. Journal of Synchrotron Radiation, 2006, 13(2): 180.
- [3] Badel X, Linnros J, Kleimann P, et al. IEEE Transactions on Nuclear Science, 2004, 51(3): 1001.
- [4] Nagarkar V V, Tipnis S V, Gaysinskiy V B, et al. International Society for Optics and Photonics, 2003, 5030: 541.
- [5] Cha B K, Bae J H, Kim B J, et al. Nuclear Inst. and Methods in Physics Research, A, 2008, 591(1): 113.
- [6] Kim B J, Cho G, Cha B K, et al. Radiation Measurements, 2007, 42(8): 1415.
- [7] Simon M, Engel K J, Menser B, et al. Medical Physics, 2008, 35(1): 968.
- [8] Derenzo S E, Moses W W. Lawrence Berkeley Laboratory Report No. LBL-33295, 1992, 1.
- [9] Luo Z P. Metallography, Microstructure, and Analysis, 2012, 1(6): 320.
- [10] Linnros J, Badel X, Kleimann P. Physica Scripta, 2006, T126: 72.
- [11] Samei E, Ranger N T, Dobbins J T, et al. Medical Physics, 2006, 33(5): 1454.
- [12] Gao P, Gu M, Liu X, et al. Applied Physics Letters, 2009, 95(22): 221904.
- [13] Cai Z, Li F, Gu M, et al. The 12th International Conference on Inorganic Scintillators and Their Applications, O2 3, Shanghai, China, 2013, 4: 15.
- [14] Cha B K, Lee S J, Muralidharan P, et al. Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment, 2011, 652(1): 717.

Fabrication and Performance of Pixelated γ -CuI Scintillation Conversion Screens Based on Oxidized Silicon Micropore Arrays

SUN Shou-qiang, GU Mu^{*}, CHEN Ting-ting, ZHANG Juan-nan, LIU Xiao-lin, LIU Bo, HUANG Shi-ming, NI Chen
Shanghai Key Laboratory of Special Artificial Microstructure Materials & Technology, School of Physics Science and Engineering, Tongji University, Shanghai 200092, China

Abstract The pixelated CuI scintillation conversion screens which have a monodisperse micro-columnar structure were fabricated with pressure-injection method using the high-purity CuI powder and oxidized silicon micropore arrays. The results of scanning electron microscopy and X-ray diffraction show that CuI micro-columns are continuous and dense with the crystal lattice of γ -phases. Their diameters, spaces and depths are about 2.5, 1.5 and 80 μm , respectively. X-ray excited luminescence (XEL) reveals that the as-prepared γ -CuI scintillation screens have an emission band near 680 nm, which has a comparatively slow decay time. After iodine-doping especially at the content of 10 Wt%, the emission band near 680 nm is effectively suppressed with an emission peak at 432 nm, which has an ultra-fast decay time. The spatial resolution of the pixelated γ -CuI scintillation conversion screens was measured with knife-edge method. The result shows that the resolution can reach to 38 lp $\cdot$ mm⁻¹, which indicate that the pixelated γ -CuI scintillation screens have excellent spatial resolution with ultra-fast time response and it possess a unique value in X-ray imaging.

Keywords Pixelated γ -CuI scintillation conversion screen; Micro-columnar structure; Scintillation property; Spatial resolution

(Received Oct. 9, 2015; accepted Mar. 16, 2016)

* Corresponding author